A Web-Based Interface for Information Propagation Analysis on Directed Acyclic Networks

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***Abstract*—**

I. Introduction

Understanding how information propagates through complex networks is fundamental to reliability engineering, risk assessment, and infrastructure resilience [2]. Network analysis allows domain experts to identify critical pathways, bottlenecks, and vulnerabilities within their systems, including the treatment of uncertainty, disruption, and unexpected threats.

Although the mathematical foundations of network analysis are well established [**?**], [1], [2], barriers remain between the availability of these methods and their routine use in practice. Domain experts often have the detailed knowledge of their systems but lack the programming skills required to effectively use many existing software approaches. This challenge is commonly described as the "last mile problem" in scientific software, where capable algorithms exist but remain inaccessible to their intended users. Studies consistently report this gap between algorithmic availability and practical adoption [**?**], [**?**].

Web-based platforms such as Jupyter notebooks [5], Shiny applications [6], and Streamlit dashboards [7] have partially addressed these challenges by lowering the learning curve through interactive interfaces and improved workflow sharing. However, these platforms remain fundamentally code-driven, requiring users to write Python, R, or similar scripting languages to define models and analyses. For decision makers without programming backgrounds, this requirement continues to limit adoption.

Visualization tools such as Gephi [8] and Cytoscape [9] have gained widespread adoption. However, these tools focus primarily on network topology and visualization. Consequently, to perform a complete analysis, users often combine them with more specialized analysis tools such as GeNIe [10], BayesiaLab [11], and Netica [**?**], which support probabilistic reasoning but generally involve a steeper learning curve.

Probabilistic network-based problems require explicit uncertainty quantification. Existing low- and no-code software tools often restrict uncertainty representation to single probability distributions or interval bounds. In networks where data are sparse, incomplete, or derived from expert judgment, uncertainty cannot always be adequately represented by a single distribution. Probability bounds analysis (PBA) and probability boxes (p-boxes) address this limitation by representing uncertainty as bounds on cumulative distributions rather than precise values [3]. Software libraries supporting p-box arithmetic exist for R [**?**], Julia [**?**], and Python [**?**]. However, users wishing to use these methods must be prepared to write code directly against these libraries.

Deployment architecture is also a significant consideration in scientific software. The local-first software movement [12] encourages the development of software that prioritizes user data sovereignty while maintaining collaborative features. Although cloud-based solutions offer convenience and scalability, they raise privacy concerns [**?**]. For industries working with proprietary infrastructure models or regulated data (such as nuclear, defence, or critical infrastructure sectors), local deployment is often a regulatory or contractual requirement rather than merely a preference.

To date, no existing tool combines (i) purely visual interaction, (ii) comprehensive network analysis capabilities, (iii) native support for uncertainty quantification using p-boxes, and (iv) guaranteed local-first deployment. This paper addresses this gap by presenting a web-based graphical interface for information propagation analysis on directed acyclic networks. The toolkit provides a fully no-code environment supporting multiple analysis modes: network structure exploration, network decomposition, and probability propagation with interval uncertainty. Built on the `InformationPropagationAnalysis.jl` Julia package [13], the system exposes these algorithms through a REST back-end server connected to an Angular-based frontend. Critically, the system operates entirely on the user's local machine, ensuring that sensitive network data never leaves the user's environment.

The decoupled frontend-backend architecture allows domain experts to use the interface without programming, while developers can extend each component independently.

The remainder of this paper is organized as follows. Section II describes the system architecture, detailing both the Julia back-end server and Angular frontend components. Section III presents the supported analysis modes and their corresponding interface elements. Section IV demonstrates the system through representative use cases. Finally, Section V concludes with a discussion of limitations and directions for future work.

II. Software Architecture

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